# Home Assignment

Q1 - CO1 :

A.

| PEAS Component | Description from Scenario |
| --- | --- |
| Performance Measure | Safe and accurate package delivery, maintaining battery efficiency, stable altitude, and avoiding collisions while completing deliveries between 9:00am - 6:00am. example, "abrupt altitude correction near Marathahalli bridge directly affects battery reserves." |
| Environment | Bangalore urban residential zones including Whitefield warehouse, customer homes, Marathahalli bridge area, tree canopies, narrow alleyways, birds, kites, hobbyist drones, varying wind conditions, and daytime weather changes. |
| Actuators | Thrust, pitch, yaw, and roll controls used by the drone to move, turn, stabilize altitude, and navigate. |
| Sensors | Down-ward facing camera, GPS, ultrasonic proximity sensors, and barometric altimeter. |

B. Observability - The drone's environment is partially observable because it cannot sense all relevant information completely.

evidence : "...cannot detect obstacles hidden under tree canopies or inside narrow alleyways."

Justification : Since the agent does not have complete access to the full state of the environment at every moment, the environment is partially observable.

Determinism - The environment is stochastic / non-deterministic because the same action may produce different outcomes under changing conditions.

evidence : "...wind gusts affect altitude differently during hot afternoons versus calm mornings."

"Birds, kites, and hobbyist drones behave unpredictably."

Justification : Even if the drone performs the same action, external conditions such as wind & moving obstacles can change the results.

| C. Property | Classification | Justification |
|---|---|---|
| • Episodic (or) Sequential | Sequential | Current decisions affect future states example, an abrupt altitude correction affects battery power. |
| • Static (or) Dynamic | Dynamic | The environment changes continuously due to moving birds, drones, kites. |
| • Discrete (or) Continuous | Continuous | Position, altitude, speed, wind variation, battery consumption, pitch, yaw, and roll change continuously over time. |
| • Single-Agent (or) Multi-Agent | Multi-Agent | Other agents like birds, kites and hobbyist drones also share the airspace and influence the drone's behavior. |

Q2 - CO1 :

A.

| PEAS Component | Identification & Justification |
| --- | --- |
| Performance Measure | The robot's goal is to perform safe, accurate, minimally invasive surgery with minimal tissue damage, controlled bleeding, and successful completion of the operation example, "an over-aggressive cut in the first 10 minutes can increase bleeding risk for the remainder of a 4-hour procedure." |
| Environment | The environment is the operation theatre at AIIMS Hyderabad, including the patient undergoing surgery, internal organs / tissues, surgical instruments, lighting conditions, human surgeon, operating staff, and continuously changing tissue conditions example, "Tissue resistance varies depending on patient age and hydration levels." |

B.

| PEAS Component | Identification & Justification |
| --- | --- |
| Actuators | The robot uses four robotic arms to manipulate surgical instruments and perform laparascopic procedures. These arms execute movements such as cutting, gripping, suturing, rotating, and positioning instruments according to the surgeon's commands. |

Sensors — The robot uses multiple sensors : 1. High-resolution endoscopic camera → provides visual information from inside the body. 2. Force-feedback sensors → measure tissue resistance & pressure applied at instrument tips. 3. Pre-loaded patient anatomy models → provides anatomical reference information for guidance during surgery.

c.

| Environment Property | Classification | Justification |
|---|---|---|
| Observability | Partially Observable | The robot cannot detect tissue micro-tears smaller than 0.2mm. ⇒ Doesn't complete information about the environment. |
| Determinism | Stochastic (or) Non-Deterministic | Outcomes are uncertain because tissue resistance changes with patient age and hydration levels. |
| Episodic (or) Sequential | Sequential | Early actions affect later outcomes ↳ an over aggressive cut early in surgery increases bleeding risk |
| Static (or) Dynamic | Dynamic | The environment changes continuously during surgery due to bleeding, tissue movement, changing resistance. |
| Discrete (or) Continuous | Continuous | Movements of robotic arms, applied force, tissue resistance, & surgical positioning vary continuously over time. |
| Single Agent (or) Multi Agent | Multi-Agent | Robot operates alongside a human surgeon who provides real-time verbal commands, making it an multi-agent environment. |

Q3-CO1 :

A. PEAS Component — Identification & Justification

| PEAS Component | Identification & Justification |
|---|---|
| Performance Measure | The robot's main performance measure are accurate medicine dispensing, speed, efficiency and safe handling of medications. The robot must "pick, verify, and deliver prescribed medications" correctly while meeting the "3 minute SLA per order." |
| Environment | The environment is the temperature-controlled pharmacy store at Yashoda Hospitals, Hyderabad. It includes 500+ medicine shelf slots, 30 dispensing counters, pharmacist staff, patient attendants, prescription orders, refrigerated storage sections, and different medication types such as tablets, syringes and liquids. |

B. PEAS Component — Identification & Justification

| PEAS Component | Identification & Justification |
|---|---|
| Actuators | The robot uses robotic picking and delivery mechanisms to perform actions such as selecting medicines from shelves, gripping packages, moving items, sorting prescriptions, and delivering medicines to dispensing counters. These actuators enable handling of tablets, syringes, liquids, and refrigerated items. |

| | |
|---|---|
| Sensors | The robot uses - 1. Barcode Scanners → identify medicine packages. 2. RFID readers → verify tagged medications and inventory. 3. Top-mounted camera → assists in package recognition and self-monitoring. |

C.

| Environment Property | Classification | Justification. |
|---|---|---|
| Observability | Partially Observable | The robot cannot detect damaged barcodes or differentiate between identical drug packages with different strengths. |
| Determinism. | Stochastic (or) Non-Deterministic. | Errors may occur due to damaged labels, misplaced items, or human movement around the pharmacy. |
| Episodic (or) Sequential. | Sequential | Each dispensing action affects future operations. |
| Static (or) Dynamic | Dynamic | The environment changes continuously because pharmacists, attendants, and incoming prescriptions are constantly moving & changing. |
| Discrete (or) Continuous | Discrete | Tasks such as scanning, picking, verifying, and delivering medicines occur as separate identifiable actions |
| Single-Agent (or) Multi-Agent | Multi-Agent | The robot works alongside pharmacist staff & interacts indirectly with patient attendants. |

Q4- CO2 :

A. Uniform Cost Search (UCS)

| Step | Expanded Node | Path Cost | Frontier After Expansion. |
|---|---|---|---|
| 1 | Mumbai | 0 | Pune(149), Surat(263) |
| 2 | Pune | 149 | Surat(263), Hyderabad (709) |
| 3 | Surat | 263 | Ahmedabad(519), Hyderabad (709) |
| 4 | Ahmedabad | 519 | Hyderabad (709), Jaipur(1188) |
| 5 | Hyderabad | 709 | Jaipur(1188), Nagpur(1212) |
| 6 | Jaipur | 1188 | Nagpur(1212), Agra(1426) |
| 7 | Nagpur | 1212 | Agra(1426), Bhopal(1569), Delhi(2287) |
| 8 | Agra | 1426 | Bhopal (1569), Delhi(1632), Delhi(2287). |
| 9 | Bhopal | 1569 | Delhi (1632), Delhi(2287), Agra(1932) |
| 10 | Delhi | 1632 | Goal Reached |

Final Path ⇒ Mumbai → Surat → Ahmedabad → Jaipur → Agra → Delhi

Total Path Cost → 1632 km

Nodes Expanded ⇒ 10

B. A* Search

| Step | Expanded Node | g(n) | h(n) | f(n)=g(n)+h(n) | Frontier After Expansion |
|---|---|---|---|---|---|
| 1 | Mumbai | 0 | 1148 | 1148 | Pune(1347), Surat (1405) |
| 2 | Pune | 149 | 1198 | 1347 | Surat(1405), Hyderabad (2201). |

| | | | | | |
|---|---|---|---|---|---|
| 3 | Surat | 263 | 1142 | 1405 | Ahmedabad (1425), Hyderabad (2201) |
| 4 | Ahmedabad | 519 | 906 | 1425 | Jaipur (1456), Hyderabad (2201) |
| 5 | Jaipur | 1188 | 268 | 1456 | Agra (1625), Hyderabad (2201) |
| 6 | Agra | 1426 | 199 | 1625 | Delhi (1632), Hyderabad (2201) |
| 7 | Delhi | 1632 | 0 | 1632 | Goal Reached. |

Final Path ⇒ Mumbai → Surat → Ahmedabad → Jaipur → Agra → Delhi.

Total Path Cost ⇒ 1632 km

Nodes Expanded ⇒ 7.

C. Comparison of UCS and A*

| Feature | UCS | A* |
|---|---|---|
| Path Found | Mumbai → Surat → Ahmedabad → Jaipur → Agra → Delhi | Mumbai → Surat → Ahmedabad → Jaipur → Agra → Delhi |
| Total Cost | 1632 km | 1632 km |
| Nodes Expanded | 10 | 7 |
| Heuristic Used | No | Yes |
| Efficiency | Less Efficient | More Efficient |

A* uses the heuristic h(n), which estimates the remaining distance to Delhi. For example, Hyderabad had a very high heuristic value, so A* understood Hyderabad was unlikely to lead to the best route & delayed expanding it. Hence, A* expands fewer nodes and is more efficient while still finding the optimal shortest path.

Q5 - CO2 :

A. A* Star :

| Step | Expanded Node | g(n) | h(n) | f(n) = g(n)+h(n) | Open List after Expansion |
|---|---|---|---|---|---|
| 1 | S | 0 | 9 | 9 | A(10), B(11) |
| 2 | A | 3 | 7 | 10 | B(11), C(12), D(12) |
| 3 | B | 5 | 6 | 11 | C(12), D(10), E(20) |
| 4 | D | 7 | 3 | 10 | C(12), G(10), E(20) |
| 5 | G | 10 | 0 | 10 | Goal Reached |

| Node | g(n) | h(n) | f(n) |
|---|---|---|---|
| S | 0 | 9 | 9 |
| A | 3 | 7 | 10 |
| B | 5 | 6 | 11 |
| C | 7 | 5 | 12 |
| D(via A) | 9 | 3 | 12 |
| D (via B) | 7 | 3 | 10 |
| E | 12 | 8 | 20 |
| G | 10 | 0 | 10 |

Expansion Order : $S \to A \to B \to D \to G$

Optimal path : $S \to B \to D \to G$

Total cost : 10 minutes

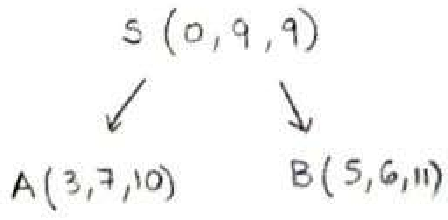
B. IDA* Search from S to G

Iteration 01 $\to$ Initial Threshold = $f(s) = 0+9 = 9$

Both $A(10)$ & $B(11) > 9 \Rightarrow$ Pruned

No goal found

New threshold = $\min(10,11) = 10$



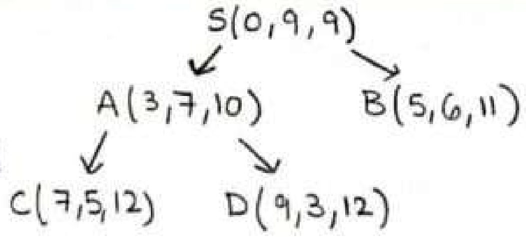
Iteration 02 → Threshold = 10

B(11) > 10 → Pruned

C(12) & D(12) > 10 → Pruned

No goal found

New threshold = min(11, 12, 12) = 11

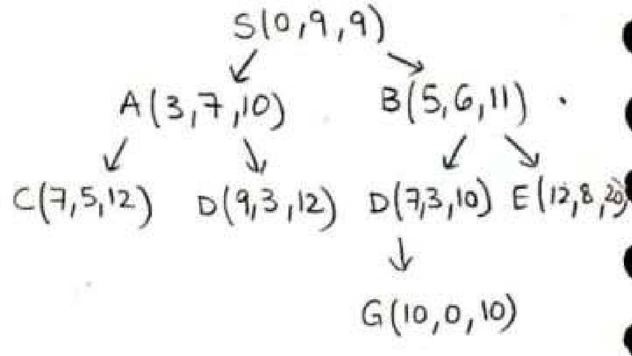



Iteration 03 → Threshold = 11

C(12) & D(12) > 11 → Pruned

E(20) > 11 → Pruned

Goal reachead at G with f = 10 ≤ 11




Threshold Sequence : 9 → 10 → 11

Final Path : S → B → D → G

Total cost : 10 minutes

C. Comparison of A* and IDA*

| Feature | A* | IDA* |
|---|---|---|
| Optimal Path | S → B → D → G | S → B → D → G |
| Total Cost | 10 minutes | 10 minutes |
| Nodes Expanded | 5 | 10 |
| Memory Requirement | Higher | Lower |
| Search Style | Best-first heuristic search | Iterative deepening heuristic search |

Q6 - CO2 :

A. A* using Manhattan Distance Heuristic

Manhattan Distance Formula,

$$h(n) = \sum |x_{current} - x_{goal}| + |y_{current} - y_{goal}|$$

| Step | State | g(n) | h(n) | f(n) = g(n) + h(n) |
|---|---|---|---|---|
| 1 | 2 8 3<br>1 6 4<br>7 - 5 | 0 | 5 | 5 |
| 2 | 2 8 3<br>1 - 4<br>7 6 5 | 1 | 4 | 5 |
| 3 | 2 - 3<br>1 8 4<br>7 6 5 | 2 | 3 | 5 |
| 4 | - 2 3<br>1 8 4<br>7 6 5 | 3 | 2 | 5 |
| 5 | 1 2 3<br>- 8 4<br>7 6 5 | 4 | 1 | 5 |
| 6 | 1 2 3<br>8 - 4<br>7 6 5 | 5 | 0 | 5 |

Optimal Cost ⇒ 5 moves

Nodes Expanded → 6

Final Path ⇒ Reached Goal Successfully.

B. A* using Misplaced Tiles Heuristic

h(n) = Number of misplaced tiles.

| Step | State | g(n) | h(n) | f(n) = g(n) + h(n) |
|---|---|---|---|---|
| 1 | 2 8 3<br>1 6 4<br>7 _ 5 | 0 | 4 | 4 |
| 2 | 2 8 3<br>1 _ 4<br>7 6 5 | 1 | 3 | 4 |
| 3 | 2 _ 3<br>1 8 4<br>7 6 5 | 2 | 3 | 5 |
| 4. | 2 8 3<br>_ 1 4<br>7 6 5 | 2 | 3 | 5 |
| 5 | _ 2 3<br>1 8 4<br>7 6 5 | 3 | 2 | 5 |
| 6 | 1 2 3<br>_ 8 4<br>7 6 5 | 4 | 1 | 5 |
| 7 | 1 2 3<br>8 _ 4<br>7 6 5 | 5 | 0 | 5 |

| Feature | Manhattan Distance | Misplaced Tiles |
|---|---|---|
| Nodes Expanded | 6 | 7 |
| Heuristic Accuracy | More informed | Less informed |
| Efficiency | Better | Lower |
| Reason | Considers exact tile distance. | Only checks whether tile is |

- Manhattan distance expands fewer nodes
  - ↳ measures how far each tile is from goal position.
  - ↳ gives more detailed guidance to A*.
  - ↳ reduces unnecessary expansions.

C. Admissibility

A heuristic is admissible if : $h(n) \leq \underline{h^*(n)}$ → true minimum remaining cost.

Manhattan distance is admissible.

↳ Each tile must move atleast its Manhattan distance to reach its goal position.
A tile cannot reach the goal in fewer moves than its row-column distance.

$$h_{Manhattan}(n) \leq \text{true cost}$$

Misplaced Tiles is admissible.

↳ Each misplaced tile requires at least one move to reach the correct position.
Therefore the heuristic never overestimates actual cost.

Consistency

A heuristic is consistent if : $h(n) \leq cost(n, n') + h(n')$

For 8-puzzle : $cost(n, n') = 1$

| Heuristic | Admissible | Consistent | More efficient |
|---|---|---|---|
| Manhattan Distance | Yes | Yes | Yes |
| Misplaced tiles | Yes | Yes | No |

Q7-CO3 :

A. CSP Formulation

1. Variables

| Variable | Department |
|---|---|
| D1 | Data Science |
| D2 | Cyber Security |
| D3 | Web Development |
| D4 | Mobile Apps |
| D5 | Clould Computing |

2. Domains

S1 → Monday Morning    S2 → Monday Afternoon

S3 → Tuesday Morning    S4 → Tuesday Afternoon.

3. Constraints

D1 ≠ D2 ; D2 ≠ D3 ; D3 ≠ D4 ; D1 ≠ D5

D4 & D5 ⇒ allowed together.

B. Variable Order : D1→D2→D3→D4→D5

Slot Order : S1 → S2 → S3 → S4

| Step | Variable Assigned | Attempted Slot | Constraint Check | Result |
|---|---|---|---|---|
| 1 | D1 | S1 | No conflict | success |
| 2 | D2 | S1 | D1 ≠ D2 × | fail |
| 3 | D2 | S2 | D1 ≠ D2 ✓ | success |
| 4 | D3 | S1 | D2 ≠ D3 ✓ | success |
| 5 | D4 | S1 | D3 ≠ D4 × | fail |
| 6 | D4 | S2 | D3 ≠ D4 ✓ | success |

| | | | | |
|---|---|---|---|---|
| 7 | D5 | S1 | D1 ≠ D5 × | fail |
| 8 | D5 | S2 | No conflict with D4 ✓ | success |

C.

| Step | Current Assignment |
|---|---|
| 1 | D1 = S1 |
| 2 | D2 = S1 × |
| 3 | D2 = S2 ✓ |
| 4 | D3 = S1 ✓ |
| 5 | D4 = S1 × |
| 6 | D4 = S2 ✓ |
| 7 | D5 = S1 × |
| 8 | D5 = S2 ✓ |

| Failed Assignment | Constraint Violated. |
|---|---|
| D2 = S1 | D1 ≠ D2 |
| D4 = S1 | D3 ≠ D4 |
| D5 = S1 | D1 ≠ D5 |

| Component | Result |
|---|---|
| CSP Variables | D1, D2, D3, D4, D5 |
| Domain | {S1, S2, S3, S4} |
| Technique Used | Backtracking Search |
| Final Schedule | D1 = S1, D2 = S2, D3 = S1, D4 = S2, D5 = S2 |

Q8 - CO3 :

A. 1. Variables

R1 → Sensor position in Row 1.

R2 → Sensor position in Row 2.

R3 → Sensor position in Row 3.

R4 → Sensor position in Row 4.

R5 → Sensor position in Row 5.

2. Domain

$R1 = R2 = R3 = R4 = R5 = \{1, 2, 3, 4, 5\}$

3. Constraints

- No two sensors can be placed in the same column.
- Two sensors cannot lie on the same diagonal.

B. State - Space Representation :

| Component | Description. |
|---|---|
| Initial State | Empty board with no sensors placed. |
| Goal State. | All 5 sensors placed without column or diagonal conflicts. |
| Operators (or) Actions | Place one sensor in a valid column of the next row |
| State Transition. | Moving from one partial placement to another by adding a new valid sensor position. |

C. Backtracking Search Tree

| Step | Assignment Attempt | Conflict Check | Result |
|---|---|---|---|
| 1 | R1 = 1 | No conflict | success |
| 2 | R2 = 1 | Same column as R1 | fail |
| 3 | R2 = 2 | diagonal with R1 | fail |
| 4 | R2 = 3 | No conflict | success |
| 5 | R3 = 1 | Same column as R1 | fail |
| 6 | R3 = 2 | diagonal with R2 | fail |
| 7 | R3 = 3 | same column as R2 | fail |
| 8 | R3 = 4 | diagonal with R2 | fail |
| 9 | R3 = 5 | No conflict | Success |
| 10 | R4 = 1 | Same column as R1 | fail |
| 11 | R4 = 2 | No conflict | success. |
| 12 | R5 = 1 | same column as R1 | fail |
| 13 | R5 = 2 | Same column as R4 | fail |
| 14 | R5 = 3 | No conflict | success |

Final 5 x 5 Sensor Grid

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | Sensor | – | – | – | – |
| 2 | – | – | Sensor | – | – |
| 3 | – | – | – | – | Sensor |
| 4 | – | Sensor | – | – | – |
| 5 | – | – | – | Sensor | – |

Q9 - CO3 :

Q

A

A. 1. Variables

Each unique letter is treated as a CSP variable.

Variable Set $\Rightarrow \{C, R, O, S, A, D, G, N, E\}$

2. Domains

Each variable can take a digit from 0 - 9.

Domain $\Rightarrow \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

3. Constraints

- Leading letters cannot be zero

  $C, R, D \neq 0$

- All letters must have different digits.

- Introduce carry variables : $c_1, c_2, c_3, c_4, c_5$.

$$
\begin{array}{ccccccc}
 & c_5 & c_4 & c_3 & c_2 & c_1 & \\
 & C & R & O & S & S \\
+ & & R & O & A & D & S \\
\hline
D & A & N & G & E & R
\end{array}
$$

Units column $\rightarrow$ $S + S = R + 10c_1$

Tens column $\rightarrow$ $S + D + c_1 = E + 10c_2$

Hundred Column $\rightarrow$ $O + A + c_2 = G + 10c_3$

Thousands Column $\rightarrow$ $R + O + c_3 = N + 10c_4$

Ten-Thousands Column $\rightarrow$ $C + R + c_4 = A + 10c_5$

Hundred-Thousands Column $\rightarrow$ $c_5 = D \Rightarrow D = 1$

B. State-Space Representation.

| Component | Description |
|---|---|
| Initial State | No letters assigned any digit. |
| Goal State | All letters assigned unique digits satisfying the arithmetic equation |

- Operators/Actions — Assign a valid unused digit to an unassigned letter.
- State Transition — Move from partial assignment to extended assignment after assigning a new digit.

C.

| Step | Assignment | Constraint Applied | Result |
|---|---|---|---|
| 1 | D = 1 | Leading digit of DANGER | success |
| 2 | S = 8 | Choose unused digit | success |
| 3 | Unit column : 8+8 = 16 | $R = 6$, $c_1 = 1$ | success |
| 4 | Tens column: 8+1+1=10 | $E = 0$, $c_2 = 1$ | success |
| 5 | try O = 2 | unused digit | tentative |
| 6 | try A = 3 | Hundreds : 2+3+1 = 6 | But G = 6 conflicts with R = 6 |
| 7 | Backtrack on A=3 | Constraint Violation | Backtrack |
| 8 | try A = 4 | Hundreds: 2+4+1 = 7 | Therefore G = 7 |

Forward Checking ⇒ After assigning R = 6, digit 6 removed from domains of all remaining variables.

Constraint Propagation ⇒ From $S+S = R + 10c_1$, assigning S immediately determines R and carry $c_1$

Early Pruning ⇒ A = 3 rejected immediately because G becomes 6 which conflicts with R.